Harmonic Weighting for Variance-Reduced Position Bias Estimation via Intervention Harvesting

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Abstract

Presentation bias remains a major challenge in learning-to-rank from implicit feedback, as it obscures the true relevance signal. We extend the intervention harvesting estimator of Agarwal et al. [1]—which estimates position bias from naturally occurring ranking variations without intrusive randomization—with a single principled modification: replacing their weighting scheme with *harmonic weighting*. This drop-in replacement retains the same ratio-of-expectations property (the sense in which the original estimator is "ratio-unbiased") while reducing the delta-method variance surrogate $V_k + V_{k'}$ unconditionally and the empirical MSE substantially in tested regimes. We establish an exact Cauchy–Schwarz guarantee that harmonic always reduces the delta-method variance surrogate $V_k + V_{k'}$, prove that harmonic is variance-optimal in the symmetric setting ($\alpha = \beta$), and derive the exact oracle optimum for the asymmetric case—which motivates harmonic as a parameter-free approximation and recovers it exactly when $\alpha = \beta$. Empirically, we observe variance reductions up to ~55% in synthetic imbalanced regimes in our tested grid, mirrored on semi-synthetic data that uses real Yahoo LTR relevance labels with simulated clicks. In one tested configuration (ratio 50:1, traffic split 80/20) harmonic achieves the same MSE as original with roughly half the data; we report this as a single data-efficiency point, not a global claim. We also show that original, min, and harmonic exhibit comparable bias under tested Position-Based Model violations, making lower-variance estimation directly advantageous for mean squared error.

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1 Introduction

Position bias—where higher-ranked results receive disproportionate user attention—systematically corrupts click signals used for learning-to-rank [9]. For instance, in e-commerce platforms a product at position 1 may receive 5–10× more clicks than an equally relevant item at position 10, creating a feedback loop that perpetuates suboptimal rankings. Unbiased learning requires accurate propensity estimates, but traditional approaches rely on costly randomized interventions (swap experiments, result randomization) that degrade user experience and cause revenue loss in production systems [9, 15].

Recent advances in counterfactual learning-to-rank have shown that unbiased learning is possible when examination propensities are known [9]. Estimating these propensities accurately remains challenging: traditional approaches rely on explicit randomized interventions that degrade user experience [15].

Intervention harvesting [1] avoids explicit randomization by leveraging naturally occurring ranking variations across concurrently deployed rankers. However, their weighting scheme can produce high estimator variance when impression counts are imbalanced — a common situation in production traffic where some documents accumulate disproportionate traffic at one position relative to another (e.g. anchor items in A/B-tested rankers).

This paper directly extends Agarwal et al. [1]. We keep their entire framework—PBM model, interventional sets, ratio estimator, AdjacentChain aggregation—and propose a single modification: replacing their weighting scheme with harmonic weighting. Key contributions: (1) any count-dependent weight preserves ratio-unbiasedness; (2) an exact Cauchy–Schwarz guarantee that harmonic always reduces the variance surrogate $V_k + V_{k'}$, yielding lower approximate ratio variance under the delta-method expansion; (3) exact optimality when propensities are equal; (4) up to 55% variance reduction in synthetic and semi-synthetic (Yahoo LTR) experiments; (5) harmonic requires roughly **half the data** to match the original estimator's accuracy.

2 Background and Preliminaries

2.1 Position-Based Propensity Model (PBM)

The Position-Based Model recognizes that higher-ranked results are more likely to be examined by users than lower-ranked results. For a query q where document d is displayed at position k, let C be the random variable indicating whether the user clicks on d, and E be the random variable denoting whether the user examines position k.

Following [1], we denote the relevance of a document as a non-random function $\mathrm{rel}(q, d)$ of q, where $\mathrm{rel}(q, d) = 1$ indicates relevant and $\mathrm{rel}(q, d) = 0$ indicates non-relevant. According to the PBM [5, 12], the probability of a click is:

$$\Pr(C = 1|q, d, k) = \Pr(E = 1|k) \cdot \mathrm{rel}(q, d) = p_k \cdot \mathrm{rel}(q, d) \qquad (1)$$

where $p_k := \Pr(E = 1|k)$ is the examination probability, which depends only on the position k.

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2.2 Interventional Sets

For each pair of positions $k \neq k' \in [M]$ (where M is the number of positions of interest), [1] defined interventional sets:

$$\begin{aligned} S_{k,k'} := \{(q,d) : q \in Q, d \in \Omega(q), \\ \exists f, f' \text{ s.t. } \mathrm{rk}(d|f(q)) = k, \\ \mathrm{rk}(d|f'(q)) = k'\} \end{aligned}$$

Each interventional set $S_{k,k'}$ contains query-document pairs where the document appeared at position k in one ranking and position k' in another.

For each $(q,d) \in S_{k,k'}$, let N_k^i and $N_{k'}^i$ denote the total impression counts at positions k and k' respectively, and let $C_k^i, C_{k'}^i$ denote the total click counts at each position.

2.3 Ratio Estimator

Agarwal et al. [1] propose the following weighted ratio estimator:

$$\hat{R}_{k,k'} = \frac{A}{B} = \frac{\sum_i \omega_i \cdot C_k^i / N_k^i}{\sum_i \omega_i \cdot C_{k'}^i / N_{k'}^i} \tag{2}$$

where $\omega_i \geq 0$ may depend on observation counts $N_k^i, N_{k'}^i$ but not on clicks. We refer to $\omega_i = c$ (any positive constant; the constant cancels in the ratio) as the *original / uniform* weighting. We adopt this estimator family exactly and analyse alternative choices of ω_i.

3 Harmonic Weighting

3.1 Weighting Schemes

We consider the following weighting schemes, all of which use weights ω_i depending only on the observation counts:

- **Original / uniform**: $\omega_i = 1$ (constant), the baseline of [1]
- **Min-weighting**: $\omega_i = \min(N_k^i, N_{k'}^i)$
- **Harmonic weighting** (our proposal):

$$\omega_i^h = \frac{N_k^i \cdot N_{k'}^i}{N_k^i + N_{k'}^i} \tag{3}$$

The harmonic weight ω_i^h is the harmonic mean of N_k^i and $N_{k'}^i$ up to a factor of 1/2. It balances the contribution of each document pair by downweighting pairs with extreme impression imbalances.

We also compare against **clipped weighting** (subsampling the majority position when $N_k^i / N_{k'}^i > \tau$, [3]). Clipping is *not* a count-only reweighting in the sense of Theorem 1: it modifies the impressions N^i and clicks C^i themselves by discarding a position-asymmetric subset, breaking the per-position binomial structure that the theorem relies on. We include it as a biased lower-variance reference, not as a counterexample to Theorem 1.

3.2 Ratio-Unbiasedness

The following theorem shows that any weight function depending only on observation counts preserves the *ratio-of-expectations* equality $\mathbb{E}[A]/\mathbb{E}[B] = p_k/p_{k'}$. We emphasise up-front that this is **not** the same as the unbiasedness of the random estimator $\hat{R}_{k,k'} = A/B$: in general $\mathbb{E}[A/B] \neq \mathbb{E}[A]/\mathbb{E}[B]$, and the gap is exactly the bias of $\hat{R}_{k,k'}$ as a random ratio. Following [1] we use "ratio-unbiasedness" as shorthand for the ratio-of-expectations property; it justifies the estimator asymptotically (since the random ratio converges to the deterministic one as the interventional set grows) but does not give an unbiased finite-sample estimator.

Theorem 1 (Ratio-unbiasedness, in the ratio-of-expectations sense). *Under the Position-Based Model, for any weights $\omega_i = \omega(N_k^i, N_{k'}^i) \geq 0$ that depend only on the observation counts (not on clicks), the ratio estimator satisfies:*

$$\frac{\mathbb{E}[A]}{\mathbb{E}[B]} = \frac{p_k}{p_{k'}}$$

provided $\mathbb{E}[B] \neq 0$. Equivalently, $\hat{R}_{k,k'}$ converges in probability to $p_k/p_{k'}$ as the interventional set $|S_{k,k'}|$ grows under standard regularity conditions.

Proof. Since ω_i is independent of C_k^i given $N_k^i, N_{k'}^i$, conditioning on counts:

$$\mathbb{E}[A] = \mathbb{E}\left[\sum_i \omega_i \frac{C_k^i}{N_k^i}\right] = \sum_{(q,d)} \mathbb{E}\left[\omega_i \cdot \frac{\mathbb{E}[C_k^i | N_k^i, N_{k'}^i]}{N_k^i}\right]$$

Under PBM, $\mathbb{E}[C_k^i | N_k^i, N_{k'}^i] = N_k^i \cdot p_k \cdot \mathrm{rel}(q,d)$, so the N_k^i terms cancel:

$$\mathbb{E}[A] = p_k \sum_{(q,d)} \mathrm{rel}(q,d) \cdot \mathbb{E}[\omega_i]$$

By the same argument, $\mathbb{E}[B] = p_{k'} \sum_{(q,d)} \mathrm{rel}(q,d) \cdot \mathbb{E}[\omega_i]$.

Taking the ratio and canceling the common factor (which is positive) yields $\mathbb{E}[A]/\mathbb{E}[B] = p_k/p_{k'}$. □

The theorem applies to original, min, and harmonic weighting. Clipped weighting falls outside its scope: clipping subsamples the impression set asymmetrically, so the resulting C_k^i, N_k^i no longer carry the per-position binomial structure on which the proof depends. (Concretely, the proof step $\mathbb{E}[C_k^i \mid N_k^i, N_{k'}^i] = N_k^i p_k \mathrm{rel}(q,d)$ would be replaced by $\mathbb{E}[\tilde{C}_k^i \mid \tilde{N}_k^i, \tilde{N}_{k'}^i] = \tilde{N}_k^i p_k \mathrm{rel}(q,d)$ for the post-clip $(\tilde{C}, \tilde{N})$, but the asymmetric selection rule makes $\tilde{N}_k^i \neq N_k^i$ in a way that does not commute with the symmetric ratio of expectations.)

4 Variance Reduction Analysis

4.1 Delta-Method Variance

We analyze the conditional variance of $\hat{R}_{k,k'} = A/B$ given observation counts $\mathbf{N}$. Since A and B are sums of independent terms (across documents), the delta method gives:

Proposition 2 (Delta-method variance). *Conditionally on observation counts $\mathbf{N} = \{N_k^i, N_{k'}^i\}$:*

$$\mathrm{Var}(\hat{R}_{k,k'} \mid \mathbf{N}) \approx R^2 \left[\alpha \cdot V_k(\omega) + \beta \cdot V_{k'}(\omega)\right] \tag{4}$$

where $R = p_k/p_{k'}$, $\alpha = (1-p_k)/p_k$, $\beta = (1-p_{k'})/p_{k'}$, and:

$$V_r(\omega) = \frac{\sum_i \omega_i^2 / N_r^i}{\left(\sum_i \omega_i\right)^2} \tag{5}$$

The terms α and β capture the binomial variance at positions k and k' respectively. Since p_k is typically larger for higher positions (higher examination probability), α tends to be smaller for positions closer to the top.

4.2 Cauchy–Schwarz Guarantee

Let $D_r = V_r(\omega^h) - V_r(1)$ denote the change in variance term at position r when switching from original to harmonic weighting.

Theorem 3 (Cauchy–Schwarz guarantee). *For any observation counts $\{N_k^i, N_{k'}^i\}_{i=1}^n$:*

$$D_k + D_{k'} \le 0$$

That is, harmonic weighting always reduces the sum of per-position variance terms compared to original weighting.

Proof. By direct computation:

$$V_r(\omega^h) = \frac{\sum_i (\omega_i^h)^2/N_r^i}{(\sum_i \omega_i^h)^2}, \quad V_r(1) = \frac{\sum_i 1/N_r^i}{n^2}$$

The sum $V_k(\omega^h) + V_{k'}(\omega^h)$ can be written as:

$$V_k + V_{k'} = \frac{\sum_i (\omega_i^h)^2(1/N_k^i + 1/N_{k'}^i)}{W^2}$$

where $W = \sum_i \omega_i^h$ and we used $\omega_i^h = \omega_i^h$ for both positions. Substituting $\omega_i^h = N_k^i N_{k'}^i/(N_k^i + N_{k'}^i)$, and noting that $(\omega_i^h)^2(1/N_k^i + 1/N_{k'}^i) = (\omega_i^h)^2 \cdot (N_k^i + N_{k'}^i)/(N_k^i N_{k'}^i) = \omega_i^h$:

$$V_k(\omega^h) + V_{k'}(\omega^h) = \frac{\sum_i \omega_i^h}{(\sum_i \omega_i^h)^2} = \frac{1}{\sum_i \omega_i^h} = \frac{1}{W}$$

For the original estimator with $\omega_i = 1$:

$$V_k(1) + V_{k'}(1) = \frac{1}{n^2}\sum_i \left(\frac{1}{N_k^i} + \frac{1}{N_{k'}^i}\right) = \frac{1}{n^2}\sum_i \frac{1}{\omega_i^h}$$

Since $W = \sum_i \omega_i^h$ and $\sum_i 1/\omega_i^h$, by the Cauchy–Schwarz inequality:

$$n^2 \le \left(\sum_i \omega_i^h\right)\left(\sum_i \frac{1}{\omega_i^h}\right) = W \cdot \sum_i \frac{1}{\omega_i^h}$$

Therefore $1/W \le (1/n^2)\sum_i 1/\omega_i^h$, i.e., $V_k(\omega^h)+V_{k'}(\omega^h) \le V_k(1)+V_{k'}(1)$, giving $D_k + D_{k'} \le 0$. □

Remark 1. *Theorem 3 is an exact, unconditional guarantee on the variance surrogate $V_k(\omega) + V_{k'}(\omega)$, holding for any impression counts. Via the delta-method approximation in Proposition 2, this translates to a reduction in $\mathrm{Var}(\hat{R} \mid \mathbf{N})$ when $\alpha \approx \beta$ (Theorem 4); for the asymmetric case, Theorem 5 provides the oracle benchmark against which harmonic is compared in Remark 2. The bound is tight: equality holds iff $N_k^i/N_{k'}^i$ is identical for all i (no heterogeneity), in which case harmonic reduces to the original estimator.*

4.3 Optimality When $\alpha = \beta$

When $\alpha = \beta$ (i.e., $p_k = p_{k'}$, as holds when adjacent propensities are close—typical under smooth position-bias decay), the conditional variance simplifies to $\alpha \cdot [V_k(\omega) + V_{k'}(\omega)]$, and Theorem 3 directly implies harmonic is optimal.

Theorem 4 (Harmonic optimality). *Among all weight functions $\omega_i = f(N_k^i, N_{k'}^i)$ with f symmetric (i.e., $f(a,b) = f(b,a)$), when $\alpha = \beta$, harmonic weighting minimizes the conditional variance* $\mathrm{Var}(\hat{R} \mid \mathbf{N})$.

Proof. When $\alpha = \beta$, the variance surrogate is proportional to $V_k(\omega) + V_{k'}(\omega) = \frac{\sum_i a_i \omega_i^2}{(\sum_i \omega_i)^2}$ with $a_i := 1/N_k^i + 1/N_{k'}^i > 0$. Set $u_i = \omega_i\sqrt{a_i}$ and $v_i = 1/\sqrt{a_i}$, so $\sum_i \omega_i = \sum_i u_i v_i$ and $\sum_i a_i \omega_i^2 = \sum_i u_i^2$. Cauchy–Schwarz gives $(\sum_i u_i v_i)^2 \le (\sum_i u_i^2)(\sum_i v_i^2)$, hence

$$\frac{\sum_i a_i \omega_i^2}{(\sum_i \omega_i)^2} \;\ge\; \frac{1}{\sum_i v_i^2} \;=\; \frac{1}{\sum_i 1/a_i},$$

with equality iff $u_i \propto v_i$, i.e., $\omega_i \propto 1/a_i$. Substituting $a_i = 1/N_k^i + 1/N_{k'}^i$ gives $\omega_i \propto N_k^i N_{k'}^i/(N_k^i + N_{k'}^i) = \omega_i^h$. Scale-invariance of the ratio estimator completes the proof. □

4.4 Oracle Asymmetric Characterization

For general $\alpha \ne \beta$, we characterize the exact oracle optimum of the surrogate $\alpha V_k(\omega) + \beta V_{k'}(\omega)$. This benchmark motivates harmonic as a practical parameter-free approximation: it is infeasible to deploy directly because α, β depend on the unknown propensities, but it provides theoretical grounding for why harmonic works well in practice.

Define $s_i = \alpha/N_k^i + \beta/N_{k'}^i$, so $\alpha V_k(\omega) + \beta V_{k'}(\omega) = \frac{\sum_i s_i \omega_i^2}{(\sum_i \omega_i)^2}$. The oracle minimiser is:

$$\omega_i^* \propto \frac{1}{s_i} = \frac{N_k^i N_{k'}^i}{\alpha N_{k'}^i + \beta N_k^i} \tag{6}$$

Theorem 5 (Exact asymmetric optimum). *For any non-negative weights ω and any fixed counts $\{N_k^i, N_{k'}^i\}$:*

$$\alpha V_k(\omega) + \beta V_{k'}(\omega) \;\ge\; \left(\sum_i s_i^{-1}\right)^{-1},$$

with equality iff $\omega_i \propto \omega_i^*$.

Proof. By Cauchy–Schwarz applied to $(\sqrt{s_i}\,\omega_i)$ and $(1/\sqrt{s_i})$: $(\sum_i \omega_i)^2 \le (\sum_i s_i \omega_i^2)(\sum_i 1/s_i)$. Dividing both sides by $(\sum_i \omega_i)^2 \cdot \sum_i 1/s_i$ gives the bound; equality holds iff $\omega_i \propto 1/s_i$. □

Corollary 6 (Unconditional improvement over original). *$\alpha V_k(\omega^*) + \beta V_{k'}(\omega^*) \le \alpha V_k(1) + \beta V_{k'}(1)$ for all counts.*

Proof. $\alpha V_k(1) + \beta V_{k'}(1) = (1/n^2)\sum_i s_i$. Cauchy–Schwarz gives $n^2 \le (\sum_i s_i)(\sum_i 1/s_i)$, hence $(1/n^2)\sum_i s_i \ge 1/\sum_i (1/s_i)$. □

Corollary 7 (Harmonic as the $\alpha = \beta$ special case). *When $\alpha = \beta$, ω_i^* reduces to harmonic weighting.*

Proof. When $\alpha = \beta$, $s_i = \alpha(N_k^i + N_{k'}^i)/(N_k^i N_{k'}^i)$, so $1/s_i \propto N_k^i N_{k'}^i/(N_k^i + N_{k'}^i) = \omega_i^h$. □

Together, Theorem 5 and its corollaries characterize the oracle: ω_i^* is the exact surrogate optimum for any (α, β) and unconditionally beats original weighting. Since ω_i^* requires the unknown propensities, it is not deployable directly. Harmonic weighting is the practical parameter-free choice: exact at $\alpha = \beta$ (Corollary 7) and with an unconditional $V_k + V_{k'}$ guarantee (Theorem 3) for any propensity values. The oracle serves as a theoretical benchmark; Remark 2 shows it tracks harmonic to within ±2% empirically.

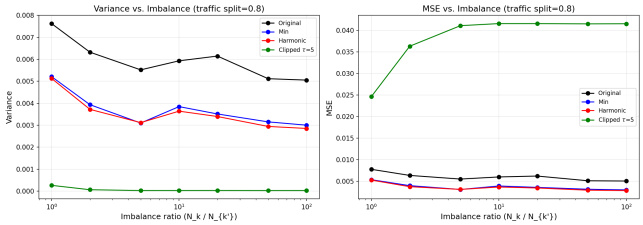

Figure 1: Variance vs. imbalance ratio at traffic split 80/20. Harmonic and min are consistently below original; clipped achieves near-zero variance at the cost of large bias^2 (see Table 1).

Remark 2 (Oracle validation). *We verified empirically that substituting the true simulation propensities into ω_i^* yields variance within ±2% of harmonic across all tested configurations. For the AdjacentChain estimator, anchor docs dominate and both weights converge to $\approx N_{k'}^i$, under severe imbalance, so the residual gap is negligible in practice.*

5 Synthetic Experiments

5.1 Experimental Setup

We evaluate all weighting schemes under the Position-Based Model with $p_k = (1/k)^\eta$, $\eta = 1.0$, $M = 10$ positions. We use the AdjacentChain estimator [1] and measure MSE, variance, and bias^2 over 500 independent trials.

5.1.1 Experiment 1: Imbalance Sweep. We simulate the anchor-item model common in production systems: a fraction s (traffic split) of documents are "anchor" items with heavy traffic at position k (impression counts drawn from Uniform$(r \cdot 50, r \cdot 200)$) and light traffic at position k' (drawn from Uniform$(1, 51)$), while the remaining $1 - s$ fraction are regular documents with balanced impressions (both positions draw from Uniform$(25, 75)$). We sweep traffic split $s \in \{0.50, 0.60, 0.70, 0.80, 0.90, 0.95\}$ and imbalance ratio $r \in \{1, 2, 5, 10, 20, 50, 100\}$.

5.1.2 Experiment 2: PBM Violations. We test robustness to model misspecification under two violation models:

- **Trust bias**: $\Pr(C = 1|q, d, k) = p_k \cdot \text{rel}(q, d) + \epsilon_k \cdot (1 - \text{rel}(q, d))$, where ϵ_k is a position-dependent trust effect [1]
- **Position-dependent relevance**: $\Pr(C = 1|q, d, k) = p_k \cdot \text{rel}(q, d, k)$ where relevance varies with position

5.2 Results: Imbalance Sweep

Figure 1 shows variance as a function of imbalance ratio at traffic split $s = 0.80$. Harmonic consistently outperforms original and min weighting, with reductions increasing as imbalance grows. Table 1 reports representative results.

Figure 2 shows variance reduction relative to original across all splits and ratios in the tested grid. At split $s = 0.80$, ratio $r = 100$, harmonic reduces variance by **43.6%** versus original and by ~5% versus min. Even at ratio $r = 1$ (no systematic imbalance) we observe a ~ 33% variance reduction because per-document count heterogeneity across queries remains. We have not encountered a tested configuration in which harmonic increases the delta-method surrogate $V_k + V_{k'}$ over original (consistent with Theorem 3), but we caution that the reported MSE depends on the full delta-method expansion — adversarial (α, β) regimes outside our tested grid are an open question. Reduction increases with traffic split, reaching ~54% at $s = 0.90$, $r = 50$. Figure 3 shows this trend across all splits. The harmonic–min gap is consistent but modest (~5–7%); the contribution of harmonic is the principled derivation (Theorem 4) and the unconditional surrogate guarantee (Theorem 3), not a large empirical margin over min.

Scheme	Var ($r = 1$)	Var ($r = 10$)	Var ($r = 100$)	MSE ($r = 100$)
Original	0.00762	0.00593	0.00505	0.00507
Min	0.00521	0.00385	0.00300	0.00302
Harmonic	0.00513	0.00364	0.00285	0.00285
Clipped$_5$	0.00026	0.000024	0.000024	0.04151

Table 1: Variance and MSE at traffic split 80/20. Harmonic consistently outperforms both original and min. Clipped achieves near-zero variance but bias^2 $\approx$ 0.041, giving MSE $\approx$ 145× higher than harmonic.

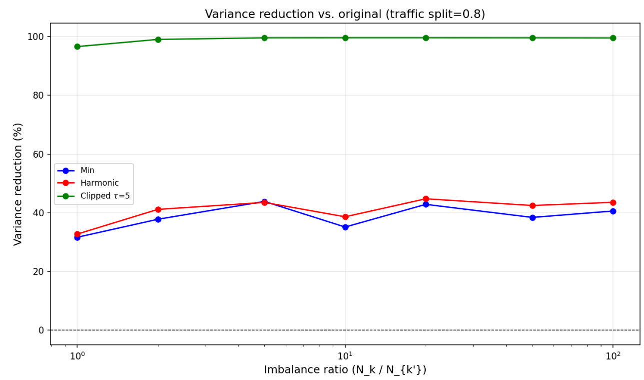

Figure 2: Variance reduction (%) of harmonic and min relative to original, across traffic splits and imbalance ratios. Harmonic consistently dominates min; gains increase with both split and ratio.

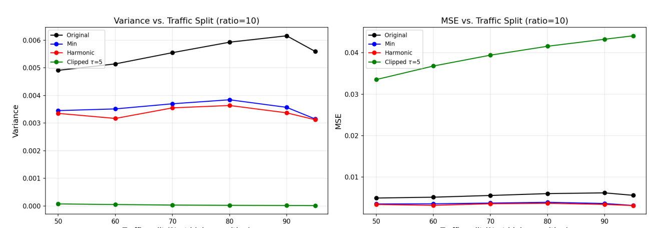

Figure 3: Variance vs. traffic split at imbalance ratio 50:1. Harmonic and min consistently outperform original across all splits; gains grow as more documents become anchor items.

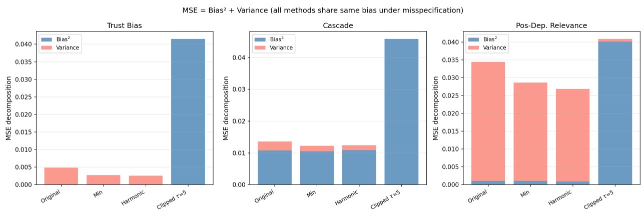

Figure 4: MSE decomposition (bias^2 + variance) under two PBM violation models. Original, min, and harmonic share comparable bias^2 under both violations; harmonic achieves lower variance and thus lower MSE. Clipped weights introduce large additional bias.

5.3 Results: PBM Violations

Figure 4 and Table 2 summarize results. Two findings stand out. First, original, min, and harmonic weighting all have *comparable bias*^{2} under each violation model, substantially lower than clipped. This is consistent with all three being ratio-unbiased under PBM: when PBM fails, the residual bias within the intervention-harvesting family is determined by the model mismatch rather than the weight choice. Second, because bias is comparable across these three schemes, the lower variance of harmonic directly translates into lower MSE: 47% lower than original under trust bias, 22% lower under position-dependent relevance.

Model	Scheme	MSE	Variance	Bias^2
Trust	Original	0.00486	0.00485	7.5e-6
	Harmonic	0.00259	0.00259	2.7e-6
	Clipped$_5$	0.04152	2.0e-5	0.04150
Pos.-dep.	Original	0.03442	0.03335	0.00107
	Harmonic	0.02684	0.02592	0.00092
	Clipped$_5$	0.04090	7.5e-4	0.04015

Table 2: MSE decomposition under two PBM violations (anchor-item imbalance, split 80/20, ratio 50:1). Original and harmonic have comparable bias^2—far below clipped—while harmonic achieves 47% lower variance under trust bias and 22% lower under position-dependent relevance. Clipped's near-zero variance comes at the cost of $160\times$ higher bias^2.

5.4 Experiment 3: Sample Efficiency

Lower variance should translate to lower data requirements. We test this on a single point of the imbalance grid (ratio 50:1, traffic split 80/20) by sweeping data volume (base impressions per document, $b \in [2, 100]$). Figure 5 shows MSE decreasing monotonically for the ratio-unbiased schemes. At $b = 5$, harmonic achieves the same MSE as original at $b \approx 10$, i.e. roughly a 2× data reduction in this single configuration. At higher data volume the relative gap widens: at $b = 50$, harmonic reduces MSE by ∼56%. Clipped weighting's MSE is flat (≈ 0.041) regardless of data volume, confirming bias-domination. We caution that the "data-efficiency factor" depends on (s, r, η) and is reported here for one cell of the grid; characterising

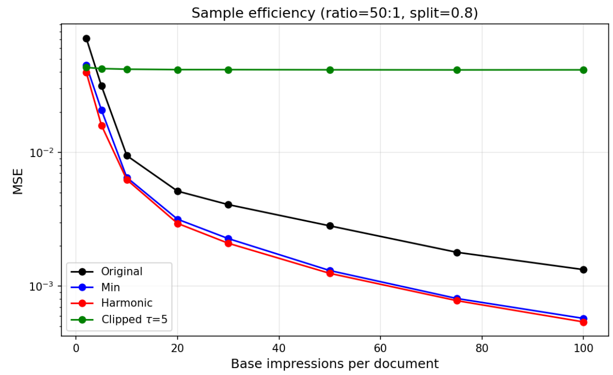

Figure 5: MSE vs. data volume (base impressions per document) at imbalance ratio 50:1, split 80/20. Harmonic reaches the same MSE as original with roughly half the data. Clipped's MSE is flat (bias-dominated) regardless of data volume.

it as a function of all three would require a heatmap-style sweep that we flag as future work.

5.5 Experiment 4: Yahoo LTR Semi-Synthetic

To verify that results hold beyond uniform-random relevance, we repeat the imbalance sweep using *real* relevance labels from the Yahoo Learning to Rank Challenge dataset (Set 1) [4]. We use the 6,936 test-set queries with graded relevance $g \in \{0, 1, 2, 3, 4\}$; continuous relevance is computed as $\mathrm{rel}(q, d) = (2^g - 1)/15$, following the standard NDCG formulation. Click probabilities follow PBM: $P(C = 1|q, d, k) = p_k \cdot \mathrm{rel}(q, d)$. All other simulation parameters (anchor-item model, $M = 10$, $\eta = 1$, 200 trials) match Experiment 1.

Scheme	Var ($r = 1$)	Var ($r = 10$)	Var ($r = 50$)	MSE ($r = 50$)
Original	0.00017	0.00015	0.00015	0.00015
Min	0.00011	0.00009	0.00009	0.00009
Harmonic	0.00010	0.00009	0.00009	0.00009
Clipped$_5$	0.00001	0.00000	0.00000	0.04120

Table 3: Variance and MSE on Yahoo semi-synthetic data (split 80/20). Harmonic consistently outperforms original and min. Clipped achieves near-zero variance but incurs large bias^2, giving MSE $\approx 458\times$ higher than harmonic at $r = 50$.

The results (Table 3 and Figure 6) show harmonic weighting reducing variance by ∼40% versus original at ratio $r = 50$, consistent with the fully synthetic results in Experiment 1. Harmonic reduces variance at every tested ratio, including $r = 1$. The relative ordering of all schemes is identical to the fully synthetic setting, suggesting that variance reduction is driven by impression imbalance and is not sensitive to the relevance distribution. **We emphasise that this experiment is semi-synthetic — relevance labels are real, but clicks are simulated under a clean PBM with the same anchor-item imbalance model as Experiment 1. It is not a real-log validation.**

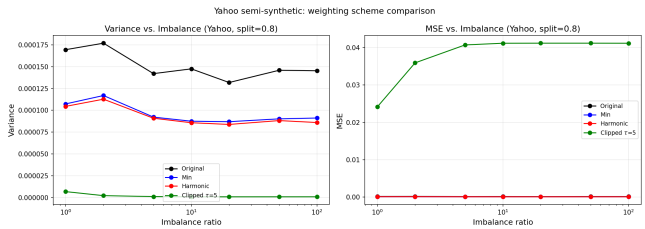

Figure 6: Variance and MSE vs. imbalance ratio on Yahoo semi-synthetic data (split 80/20). Harmonic consistently outperforms original and min across all ratios.

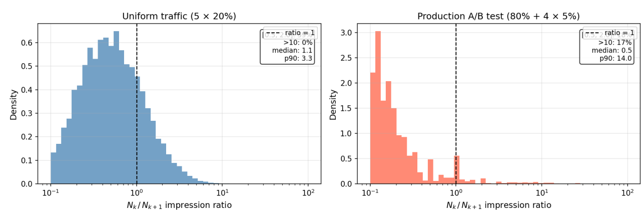

Figure 7: Impression ratio N_k/N_{k+1} under uniform vs. production traffic. Production A/B testing creates a heavy-tailed distribution (16.7% of pairs with ratio >10, p90=14); uniform traffic stays near ratio=1, consistent with Baidu-ULTR.

5.6 Experiment 5: Production-style Traffic Simulation

A natural concern is whether anchor-item impression imbalance is realistic or synthetic. We simulate $K = 5$ concurrent rankers: a default ranker (fraction α_0 of traffic) shows each document at its natural position; treatment rankers independently swap adjacent pairs with probability 0.5. We compare uniform traffic (5×20%) against a production configuration (80% + 4×5%).

Under uniform traffic, 58.9% of adjacent pairs fall in $[0.5, 2.0]$ and only 0.4% exceed 10—consistent with Baidu-ULTR (Section 5.7). Under the production configuration, only 18.5% remain in $[0.5, 2.0]$ and 16.7% exceed 10. Harmonic weighting reduces MSE by **17.2%** in the production configuration vs. 6.3% under uniform, confirming that the benefit scales directly with traffic-induced imbalance.

5.7 Public dataset analysis: Baidu-ULTR

A natural question is whether public benchmarks exhibit the impression imbalance that motivates our method. We analysed the Baidu-ULTR dataset [16] (train split, 14.5 M rows, 12.3 M unique (q, d, pos) cells), the most widely used public click log for ULTR research. Aggregating individual impressions to (q, d, pos) cells gives:

- **94.1%** of (q, d, pos) cells have $N = 1$ impression; **98.7%** have $N \leq 3$ (mean 1.16).
- Only **4.3%** of (q, d) pairs appear at two or more distinct positions.
- Among the 454 K qualifying adjacent-pair items, the impression ratio N_k/N_{k+1} is tightly concentrated near 1: **78.4%** of pairs fall within $[0.5, 2.0]$, median ratio $\approx$ 1.0, only 0.8% exceed 10.

Why this is a property of the released dataset, not of production logs. The published Baidu-ULTR release has been deduplicated and aggregated for distribution: per-row impressions are not preserved, and 94% of cells appear with $N = 1$ because the release is effectively a deduplicated query-document-position trace. A ranker deployed in production rarely shows a given (q, d) at the same position only once; production click logs without post-hoc deduplication routinely accumulate dozens to hundreds of impressions on the same (q, d, pos), with the count distribution highly imbalanced across positions because of A/B-testing traffic splits and ranker-policy heterogeneity. The conditions that make weighting matter —heterogeneous N_k^i across documents, with anchor items concentrating traffic at one position— are erased by the dataset's pre-release aggregation, not absent from the underlying click stream. This is therefore a limitation of *this public dataset as released*, not of the method's applicability to real production logs.

Implications for our analysis. Public click logs that preserve raw per-impression counts (or reasonable aggregations of them) are needed to validate harmonic on real data. We hope the present paper motivates such releases. In the meantime, the production A/B-testing simulation (Section 5.6) and the semi-synthetic Yahoo experiment (Section 5.5) are the closest available proxies: neither is a real production log, and we mark them as such throughout.

6 Related Work and Conclusion

Position bias estimation methods broadly fall into three categories: randomized interventions, joint relevance-propensity models, and intervention harvesting.

Randomized interventions explicitly manipulate result rankings to estimate examination probabilities [9, 15]. These approaches guarantee unbiasedness by randomizing item placements, yet they significantly degrade the user experience due to intrusive changes in rankings [9].

Joint relevance-propensity models simultaneously estimate relevance and propensity from observational data using techniques like Expectation-Maximization (EM) [15, 15]. Extensions such as PAL [7] integrate position-bias correction directly into CTR prediction models for recommender systems. While these methods are non-intrusive, they require a parametric relevance model; empirical comparison against intervention harvesting is therefore methodologically ill-posed, since the two paradigms solve different estimation problems (parametric relevance-and-propensity joint inference vs. model-free propensity-only estimation).

Intervention harvesting approaches, originally proposed by [1], leverage natural ranking variations from multiple historical rankers, eliminating explicit randomization. Such methods preserve user experience while producing unbiased propensity estimates without assuming a relevance model. Extensions of this idea include temporal fluctuations in e-commerce search rankings [2] and policy-aware estimation from bandit logs [11].

Despite advances, propensity estimation remains prone to high variance with importance sampling [13]. Variance-reduction strategies include clipping [3], self-normalized estimators [14], doubly robust estimators [6, 11], and control variates [10]. Doubly robust methods for LTR [11] target the ranking objective and are not directly comparable to model-free propensity estimation. Within the intervention harvesting family, applying SNIPS-style normalization to both A and B is a no-op: dividing both numerator and denominator by $\sum_i \omega_i$ leaves $\hat{R} = A/B$ unchanged; SNIPS targets absolute-quantity estimation, not ratios. Clipped weighting [3] is the natural biased low-variance reference included in our comparison.

We have presented harmonic weighting as a drop-in replacement for the weighting scheme in the intervention harvesting estimator of Agarwal et al. [1], keeping their framework—PBM model, interventional sets, ratio estimator, AdjacentChain—entirely intact. Compared to their original weighting and the min-weighting heuristic also proposed in [1], harmonic weighting has a principled theoretical foundation and consistent empirical advantages. Key contributions:

- A unified ratio estimator framework: any count-dependent weight preserves ratio-unbiasedness (Theorem 1)
- An exact Cauchy–Schwarz guarantee: harmonic always reduces the variance surrogate $V_k + V_{k'}$ (Theorem 3), yielding lower approximate ratio variance under the delta-method expansion
- Oracle asymmetric characterization: exact surrogate-optimal weight ω_i^* (Eq. 6) unconditionally beats original (Theorem 5, Corollary 6); harmonic is its α=β special case (Corollary 7), motivating it as the parameter-free practical choice
- Empirical observations of up to ~55% variance reduction versus original in our tested (s, r) grid, mirrored on Yahoo semi-synthetic data and a production-style traffic simulation; under tested PBM violations, original, min, and harmonic share comparable bias, so the lower variance translates into lower MSE within these regimes

Our implementation and all experiments presented in this paper will be publicly available after acceptance. The implementation builds upon the unbiased learning-to-rank toolkit by [8], which provides a comprehensive framework for position bias estimation and evaluation.

Several directions remain open.

(a) AdjacentChain error propagation. Our theory covers the per-pair ratio $\hat{R}_{k,k'}$. The deployed AdjacentChain estimator multiplies $M - 1$ such ratios to recover absolute propensities, and chain-length sensitivity is not directly characterised by Theorem 3. An empirical study varying chain length (positions 1–3, 1–5, 1–10) and including non-adjacent interventional sets, with MSE reported on the final propensity estimate (not just the per-pair ratio), would close this gap.

(b) Adversarial regimes for harmonic. Theorem 3 guarantees that harmonic reduces $V_k + V_{k'}$ unconditionally. The full delta-method variance is $\alpha V_k + \beta V_{k'}$, and for sufficiently asymmetric (α, β) the oracle weight ω_i^* (Eq. 6) can in principle differ materially from ω_i^h. We have not encountered a tested grid configuration where harmonic underperforms original on $\mathrm{Var}(\hat{R} \mid \mathbf{N})$ itself, but characterising the boundary explicitly — including any worst-case count configuration — remains open.

(c) Plug-in oracle. Theorem 5 motivates a two-stage adaptive scheme: use harmonic to obtain preliminary $\hat{p}_k$, compute $\hat{\alpha}, \hat{\beta}$, then form plug-in weights $\hat{\omega}_i^* \propto N_k^i N_{k'}^i / (\hat{\alpha} N_{k'}^i + \hat{\beta} N_k^i)$ on held-out data. Held-out construction preserves count-only weighting and should approach the oracle asymptotically. We have not yet implemented this; if it consistently improves over harmonic, it becomes the stronger contribution; if not, it strengthens harmonic as the parameter-free choice.

(d) Statistical reporting. Tables 1, 2, 3 report point estimates over 200–500 trials. Adding standard errors and paired confidence intervals (particularly on the harmonic–min margin) is a writing-pass priority; some 5–7% gaps may not survive significance testing.

(e) Realistic ranker overlap. Our production-style simulation (Section 5.6) uses adjacent swaps with probability 0.5. More realistic experiments would use correlated-score perturbations or learned-ranker variations; this is straightforward future work.

(f) Real production-log validation. The most important missing piece is validation on a click log that preserves per-impression counts (the deduplicated Baidu-ULTR release does not, see Section 5.7). We hope this work motivates such releases.

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